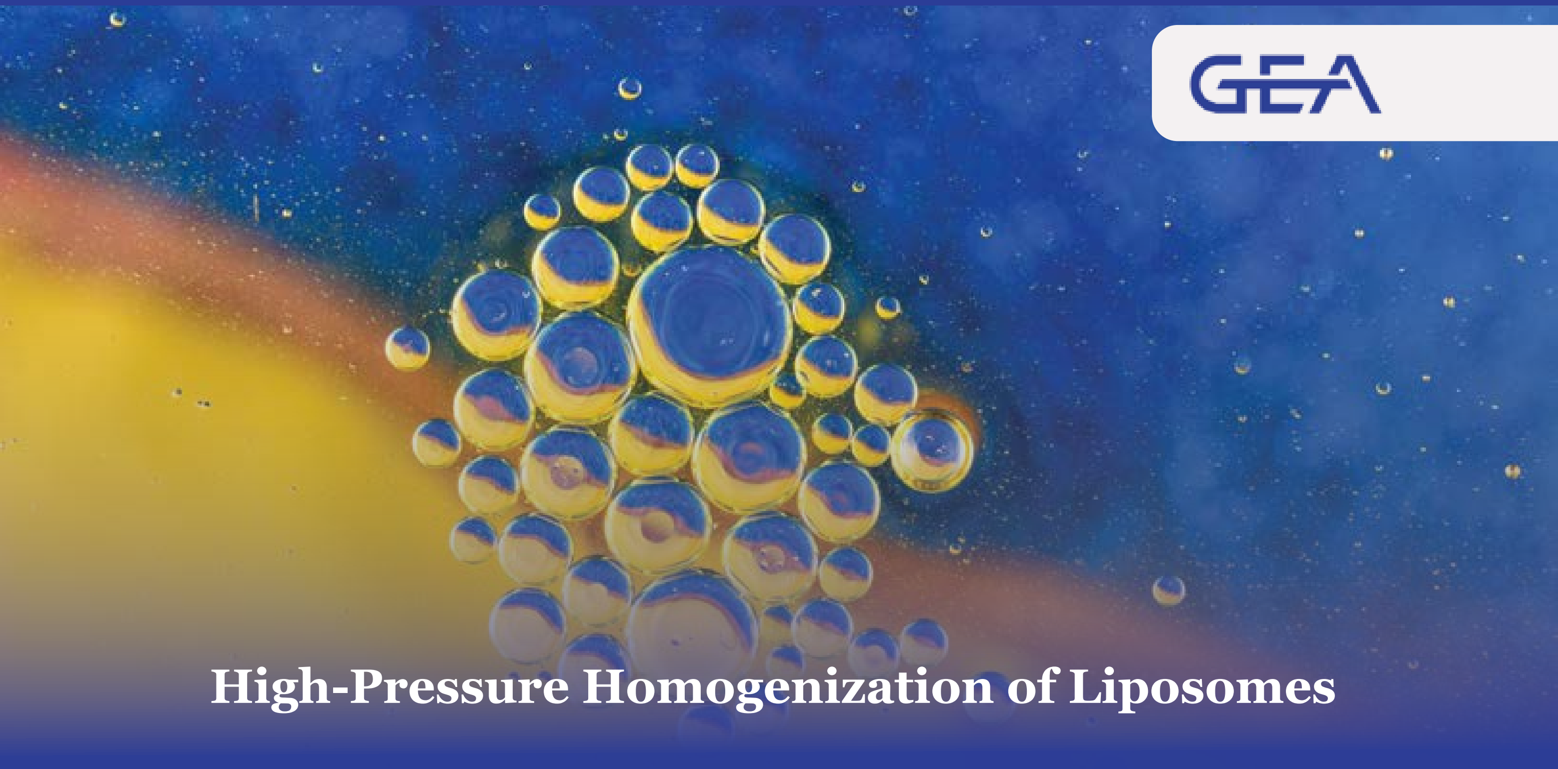

High-Pressure Homogenization of Liposomes

A Study Using the GEA PandaPLUS 2000

Homogenization plays a critical role in numerous industries. By reducing the particle size and creating a more uniform distribution, the efficiency of particles multiplies. In pharmaceutical industry, homogenization facilitates drug delivery. The reduction in particle size aids in increased bioavailability of the drug.

This study demonstrates the effective use of a GEA PandaPLUS 2000 homogenizer for reducing the particle size and providing uniform distribution of liposomes.

Keywords or phrases : Homogenization, particle size, uniform distribution, liposome, GEA PandaPLUS 2000, homogenizer.

Introduction

Liposomes have wide applications in drug delivery, cosmetics and other biomedical applications. Achieving a uniform and consistent liposome distribution is necessary for their efficiency.

Structure

Liposomes are closed, spherical shells containing aqueous solution; the shell is made by one or more lipid bilayers, and a lipid is any type of molecule soluble in no polar, organic solvents such as chloroform and ether. The type of lipid associated with liposomes is phospholipid.

The phospholipid molecule has a water-soluble end and an oil-soluble end, and an example of a phospholipid is lecithin in egg yolk.

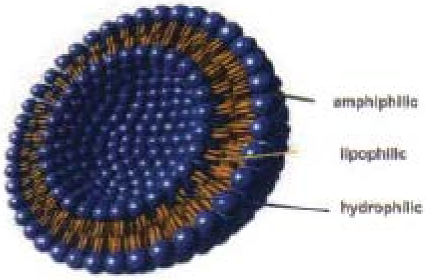

Figure 1: Structure of a liposome

The double layer is composed of two monolayers of phospholipid with the non-polar (oil-soluble) molecular ends facing inward and with the polar ends of both monolayers facing out. The liposome vesicle can be made up of a single bilayer or multiple bilayers. They are described with respect to size and number of layers (lamella):

a) 0.02 to 0.05 µm - small unilamellar vesicles
b) 0.10 to 1.00 µm - large unilamellar vesicles
c) 0.10 to 5.00 µm - multilamellar vesicles

The importance of liposomes lies in the fact that they can be used as carriers for drugs, enzymes and other biologically active molecules. These active ingredients are encapsulated in the liposome and are transported by these vesicles to different organs in the body.

Liposome vesicles can be prepared with different methods:

- Homogenisation
- French press
- Sonication by probe or cleaning bath
- Detergent dilution and dialysis
- Injection of ethanol solution into an aqueous solution

Even though liposomes are expensive to prepare, usually because of the high cost of ingredients and the small sample size (2 ml), their potential importance and usefulness have led researchers into the development of larger batches.

The Homogenizing Process

Product: Liposomes Emulsion

Composition: Water, Sodium Salicylate, Phospholipids

Application: Cosmetic and Pharmaceutical

Advantages of Homogenization: Before the treatment, the product was separated in two phases- fat and water. After the high-pressure homogenization, it was possible to obtain a stable emulsion, reducing the mean size of the fat droplets.

Homogenizer: PandaPLUS 2000

nºStage: 1

Valve 1st: "R"

Suggested Pressure: 1000 bar X2 pass

Suggested Temperature: 65-70 ºC

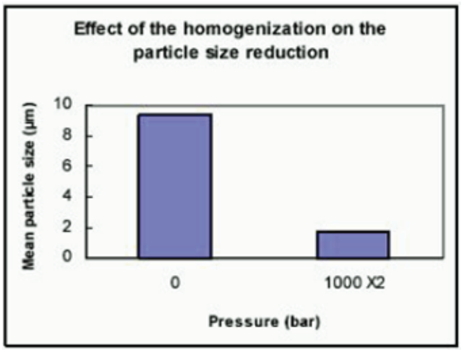

Figure 2. Effect of homogenization on particle size reduction.

In order to prepare liposomes of smaller diameter (about 110 nm), the suitable homogenising pressure is very high, about 1000-1500 bar and normally needs multiple passes (about 3) to obtain a narrow size distribution.

The results obtained can be checked utilizing different analysis technologies : electron microscopy, light scattering and dynamic light scattering such as performed with a Coulter analyser.

The homogenizer used for the treatment is a machine designed for 1200 bar for

- Very high pressure (VHP)
- High abrasion
- Pharma execution

Conclusion

This study demonstrates the successful utilisation of the GEA PandaPLUS 2000high-pressure homogenizer for the efficient and controlled reduction of liposome size. By systematically varying processing parameters, such as pressure, number of passes, and temperature, a significant decrease in liposome diameter and narrowing of liposome diameter were observed.

The Niro Soavi homogenisers suitable for very small production, can be used successfully for bigger scale preparation of liposome vesicles larger batches.

Ensuring Sterile Homogenization

Maintaining sterility is imperative to ensure the aseptic processing of materials. For this purpose, the homogenizers can be integrated with Steam Jet Sterilizers.

Unlike traditional methods that require removing parts and placing them in an autoclave, Steam Jet Sterilizers utilize high-pressure jet steam to sterilize the entire homogenizer in place.

Because of this major advantage, they are termed as SIP – Sterilization in Place.

In-place sterilization is a process that sterilizes production equipment without the need to disassemble it first, saving time.

This is typically achieved using superheated steam. Sterilization in Place systems is employed in environments where microbial contamination control is critical.

One of the main applications of SJS technology is the sterilization of homogenizers.

The optimum conditions are 121 °C, 2-3 bar for 20 mins. (conditions vary by application).

By incorporating a steam jet sterilizer into the homogenization process, it is possible to achieve both particle size reduction and complete sterility, whilst ensuring the safety of the final product.

GEA PandaPLUS 2000 in conjunction with Inkarp Steam Jet Sterilizer